PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 5+ years of experience in natural language processing (NLP) and document classification. Experienced in taking work from research and prototypes through to production models. PhD candidate researching natural language processing for Croatian.

PROFESSIONAL EXPERIENCE

ML Engineer | Slavonica Document AI d.o.o., Osijek | 03/2023–Present

- Developed an automated classification model for incoming documents (invoices, contracts and administrative decisions) for public administration clients, achieving 94% accuracy across 23 classes.
- Fine-tuned BERTić and XLM-R transformer models for named entity recognition on Croatian documents.
- Introduced MLflow for experiment tracking and model registry; deployed models as FastAPI services in Docker containers on Azure (Azure ML and App Service).
- Established production model quality monitoring for data drift, with weekly reporting.
- Mentor two students during internships.

Data Scientist | Baranjska kreditna banka d.d., Zagreb (remote from Osijek) | 09/2021–02/2023

- Developed customer churn prediction models using gradient boosting and LightGBM for the sales team.
- Analyzed customer complaint text using topic classification (TF-IDF and logistic regression, later fastText).
- Created SQL reports using an Oracle data warehouse.

Research Project Assistant | Faculty of Applied Informatics, University of Drava | 10/2020–08/2021

- Built a corpus of Croatian legal texts and annotated the data; wrote scripts in Python.

EDUCATION

PhD studies in Computer Science | University of Drava | 2022–Present
Research topic: Named entity recognition in Croatian legal texts.

Master of Computer Engineering (Mag. ing. comp.) | University of Drava | 2020

TECHNICAL SKILLS

Python, PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, pandas, NumPy, SQL, MLflow, FastAPI, Docker, Azure ML, Git, Linux

LANGUAGES

Croatian (native), English (C1), German (B2)